

Math Nuggets for GEN AI

From Vector Basics to Word Embeddings: The Mathematical Foundation of Language Understanding

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September 20, 2025

Introduction

How does a computer understand that "king" and "queen" are related? How can it know that "Paris" is to "France" as "Tokyo" is to "Japan"? The answer lies in vectors - mathematical objects that can capture relationships and meanings in ways that revolutionize how machines process language.

This note explores the journey from basic 2D vectors to sophisticated word embeddings, showing how simple mathematical concepts enable computers to understand and manipulate human language with remarkable sophistication.

1. Vector Fundamentals in 2D

1.1 What is a Vector?

A vector is a mathematical object that has both magnitude (size) and direction. Think of it as an arrow pointing from one location to another in space.

Mathematical Representation: A 2D vector $\vec{v}$ can be written as:

$$\vec{v} = \begin{bmatrix} x \\ y \end{bmatrix} = x\hat{i} + y\hat{j} \quad (1)$$

where x and y are the horizontal and vertical components, and $\hat{i}$, $\hat{j}$ are unit vectors.

Example:

$$\vec{A} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}, \quad \vec{B} = \begin{bmatrix} -2 \\ 1 \end{bmatrix} \quad (2)$$

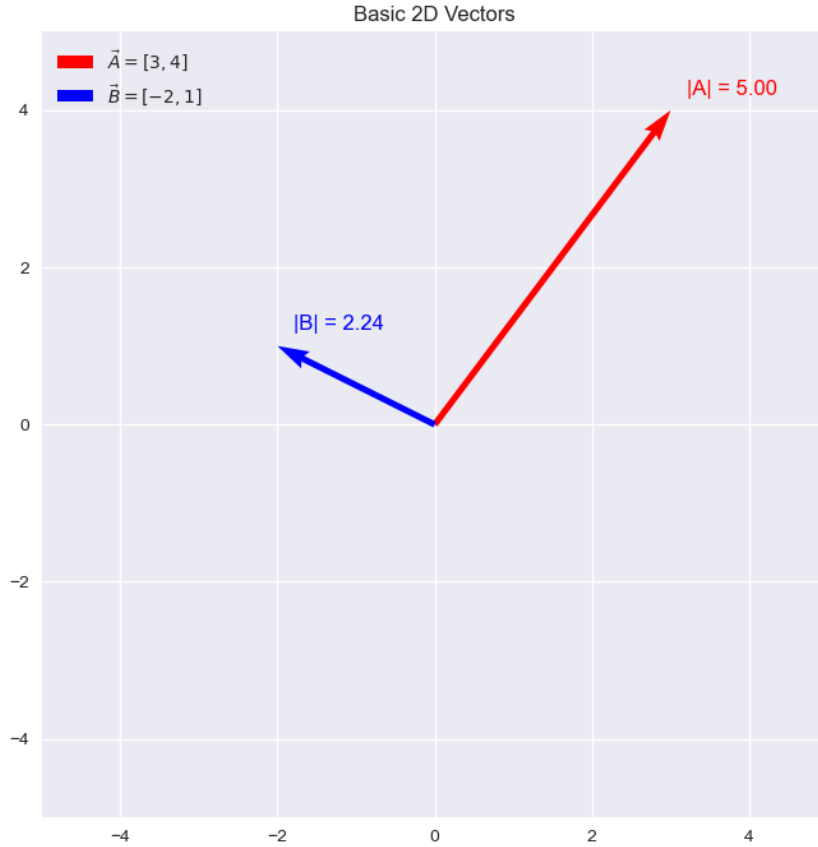


Figure 1: Basic 2D vectors showing magnitude and direction

1.2 Vector Magnitude

The magnitude (length) of a vector $\vec{v} = [x, y]$ is calculated using the Pythagorean theorem:

$$|\vec{v}| = \sqrt{x^2 + y^2} \quad (3)$$

Example: For $\vec{A} = [3, 4]$:

$$|\vec{A}| = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5 \quad (4)$$

2. Vector Operations: Dot Product and Cross Product

2.1 Dot Product: Measuring Similarity

The dot product measures how much two vectors point in the same direction. It's fundamental to understanding similarity in machine learning.

Mathematical Definition: For vectors $\vec{A} = [a_1, a_2]$ and $\vec{B} = [b_1, b_2]$:

$$\vec{A} \cdot \vec{B} = a_1b_1 + a_2b_2 = |\vec{A}||\vec{B}| \cos \theta \quad (5)$$

where θ is the angle between the vectors.

Step-by-Step Example: Given $\vec{A} = [3, 4]$ and $\vec{B} = [1, 2]$:

$$\vec{A} \cdot \vec{B} = (3)(1) + (4)(2) = 3 + 8 = 11 \quad (6)$$

$$|\vec{A}| = \sqrt{3^2 + 4^2} = 5 \quad (7)$$

$$|\vec{B}| = \sqrt{1^2 + 2^2} = \sqrt{5} \quad (8)$$

$$\cos \theta = \frac{11}{5\sqrt{5}} = \frac{11}{5\sqrt{5}} \approx 0.983 \quad (9)$$

Since $\cos \theta$ is close to 1, the vectors point in similar directions.



(a) Vectors and angle visualization

Dot Product Calculation

Vector A = [3 2]
Vector B = [2 3]

Dot Product = A · B = 12

(b) Step-by-step calculation

Figure 2: Dot product visualization showing angle between vectors and mathematical computation

2.2 Cross Product: Measuring Perpendicularity

In 2D, the cross product gives us the area of the parallelogram formed by two vectors. Using matrix determinant notation:

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & 0 \\ b_1 & b_2 & 0 \end{vmatrix} = (a_1b_2 - a_2b_1)\hat{k} \quad (10)$$

Step-by-Step Matrix Calculation: For $\vec{A} = [3, 4]$ and $\vec{B} = [1, 2]$:

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 4 & 0 \\ 1 & 2 & 0 \end{vmatrix} \quad (11)$$

Expanding the determinant:

$$= \hat{i}(4 \cdot 0 - 0 \cdot 2) - \hat{j}(3 \cdot 0 - 0 \cdot 1) + \hat{k}(3 \cdot 2 - 4 \cdot 1) \quad (12)$$

$$= 0\hat{i} - 0\hat{j} + (6 - 4)\hat{k} \quad (13)$$

$$= 2\hat{k} \quad (14)$$

The magnitude tells us the area (2 units²), and the direction ($\hat{k}$) indicates the parallelogram orientation.

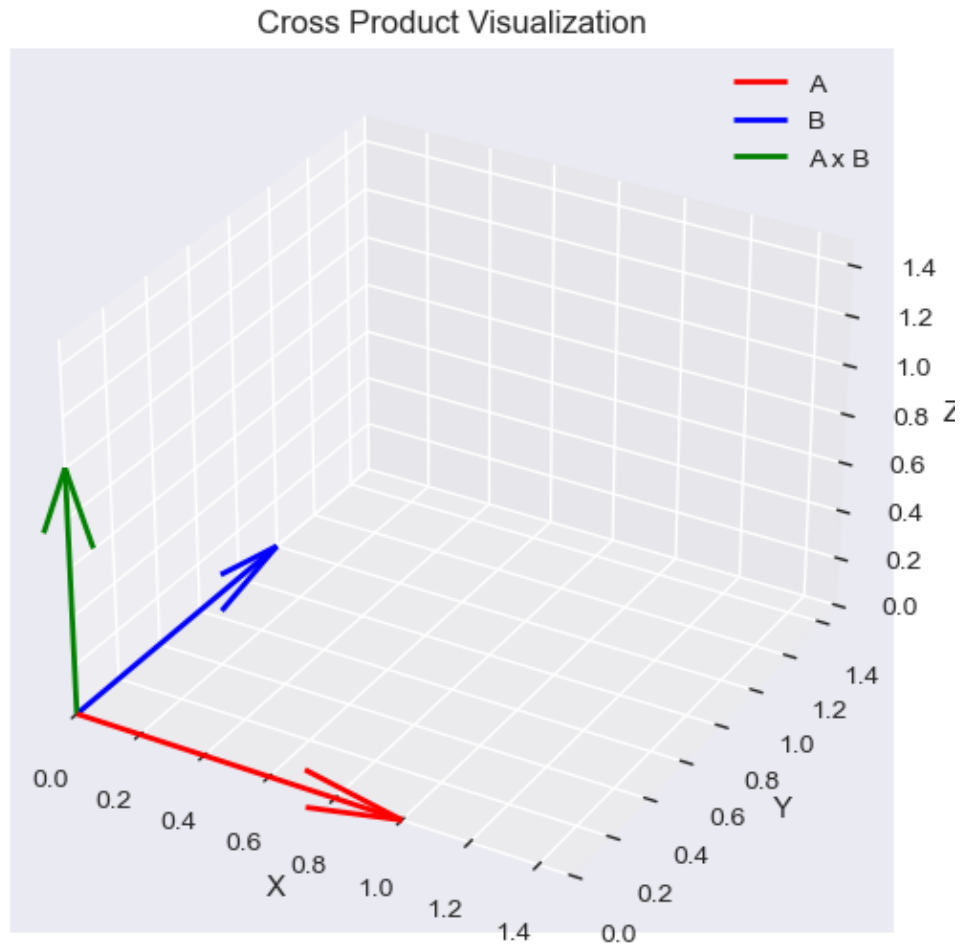


Figure 3: Cross product showing parallelogram area and matrix determinant calculation

3. Distance Between Vectors: Cosine Similarity

3.1 Why Cosine Distance Matters

In high-dimensional spaces (like word embeddings), we care more about direction than magnitude. Cosine similarity focuses purely on the angle between vectors.

Cosine Similarity Formula:

$$\text{cosine_similarity}(\vec{A}, \vec{B}) = \frac{\vec{A} \cdot \vec{B}}{|\vec{A}| |\vec{B}|} = \cos \theta \quad (15)$$

Cosine Distance:

$$\text{cosine_distance}(\vec{A}, \vec{B}) = 1 - \text{cosine_similarity}(\vec{A}, \vec{B}) \quad (16)$$

3.2 Practical Example with Distance Table

Consider three vectors representing different concepts:

$$\vec{v}_1 = [4, 3] \quad (\text{Similar direction to } \vec{A}) \quad (17)$$

$$\vec{v}_2 = [1, 4] \quad (\text{Different direction}) \quad (18)$$

$$\vec{v}_3 = [-3, -4] \quad (\text{Opposite direction}) \quad (19)$$

Computing similarities and distances with $\vec{A} = [3, 4]$:

Vector Pair	Cosine Similarity	Cosine Distance	Interpretation
$\vec{A}$ vs $\vec{v}_1$	0.96	0.04	Very Similar
$\vec{A}$ vs $\vec{v}_2$	0.92	0.08	Similar
$\vec{A}$ vs $\vec{v}_3$	-1.00	2.00	Opposite

Table 1: Cosine similarity and distance comparison

Detailed Calculations:

$$\cos \theta_1 = \frac{[3, 4] \cdot [4, 3]}{5 \cdot 5} = \frac{24}{25} = 0.96 \quad (20)$$

$$\cos \theta_2 = \frac{[3, 4] \cdot [1, 4]}{5 \cdot \sqrt{17}} = \frac{19}{5\sqrt{17}} \approx 0.92 \quad (21)$$

$$\cos \theta_3 = \frac{[3, 4] \cdot [-3, -4]}{5 \cdot 5} = \frac{-25}{25} = -1.0 \quad (22)$$

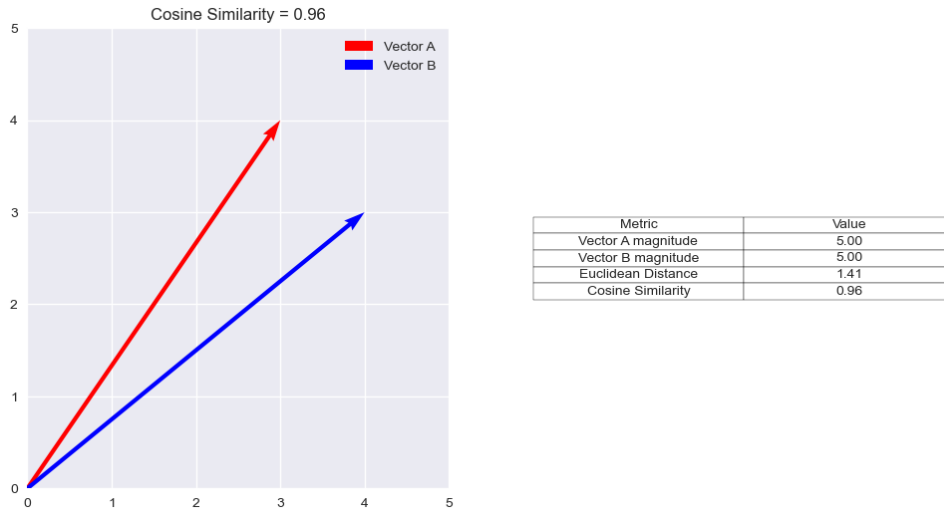


Figure 4: Cosine similarity comparison between different vector pairs

4. Introduction to Word Embeddings

4.1 The Concept of Embeddings

Word embeddings are dense vector representations of words where semantic relationships are captured as geometric relationships in vector space.

Key Principle: Words with similar meanings should have similar vector representations (high cosine similarity).

From Text to Vectors:

- Each word gets mapped to a high-dimensional vector (typically 100-1000 dimensions)
- Similar words cluster together in this space
- Relationships become vector arithmetic operations

4.2 Why Embeddings Work

Distributional Hypothesis: "Words that appear in similar contexts have similar meanings."

If we see:

- "The king ruled the kingdom"
- "The queen ruled the kingdom"

The model learns that "king" and "queen" are similar because they appear in similar contexts.

4.3 Mathematical Framework

For a vocabulary of size V , each word w_i is represented as:

$$\vec{e}_{w_i} \in \mathbb{R}^d \quad (23)$$

where d is the embedding dimension (e.g., 300).

The embedding matrix is:

$$E = \begin{bmatrix} \vec{e}_{w_1}^T \\ \vec{e}_{w_2}^T \\ \vdots \\ \vec{e}_{w_V}^T \end{bmatrix} \in \mathbb{R}^{V \times d} \quad (24)$$

5. The Famous King-Queen Example in 2D

5.1 Vector Arithmetic in Embedding Space

One of the most celebrated properties of word embeddings is that semantic relationships can be expressed as vector arithmetic:

$$\vec{king} - \vec{man} + \vec{woman} \approx \vec{queen} \quad (25)$$

5.2 Simulated 2D Example

Let's create a simplified 2D representation to visualize this concept:

$$\vec{king} = [0.8, 0.6] \quad (\text{Royal} + \text{Male}) \quad (26)$$

$$\vec{queen} = [0.7, -0.5] \quad (\text{Royal} + \text{Female}) \quad (27)$$

$$\vec{man} = [-0.2, 0.9] \quad (\text{Male}) \quad (28)$$

$$\vec{woman} = [-0.3, -0.8] \quad (\text{Female}) \quad (29)$$

Step-by-Step Calculation:

$$\vec{result} = \vec{king} - \vec{man} + \vec{woman} \quad (30)$$

$$= [0.8, 0.6] - [-0.2, 0.9] + [-0.3, -0.8] \quad (31)$$

$$= [0.8 + 0.2 - 0.3, 0.6 - 0.9 - 0.8] \quad (32)$$

$$= [0.7, -1.1] \quad (33)$$

Cosine Similarity with Queen:

$$\text{similarity} = \frac{[0.7, -1.1] \cdot [0.7, -0.5]}{||[0.7, -1.1]|| \cdot ||[0.7, -0.5]||} = \frac{0.49 + 0.55}{\sqrt{1.7} \cdot \sqrt{0.74}} \approx 0.93 \quad (34)$$

This high similarity confirms that our vector arithmetic successfully captures the gender-royalty relationship!

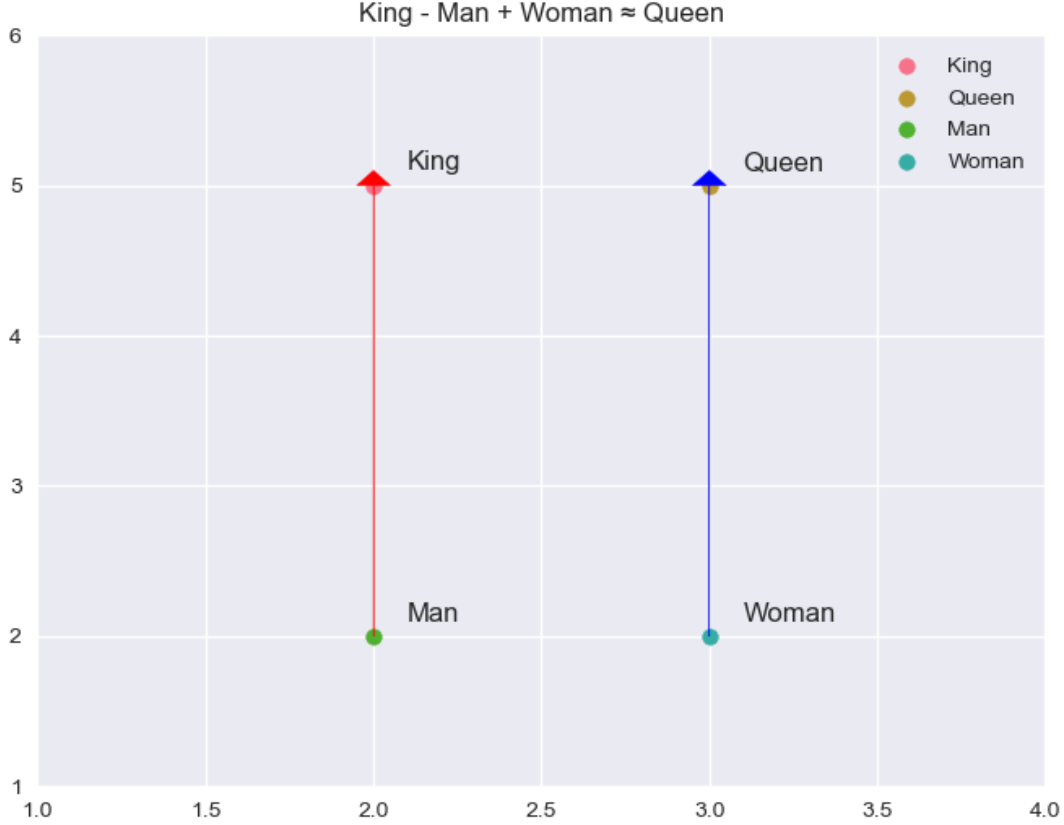


Figure 5: 2D visualization of the King-Queen analogy showing vector arithmetic

5.3 Understanding the Mathematics

The success of this analogy suggests that the embedding space captures two independent dimensions:

- **Gender dimension:** Difference between male and female concepts
- **Royalty dimension:** Difference between royal and common concepts

$$\vec{king} - \vec{man} = \text{royalty vector} \quad (35)$$

$$\vec{woman} + \text{royalty vector} = \vec{queen} \quad (36)$$

6. Real-World Applications and Extensions

6.1 Beyond Simple Analogies

Word embeddings enable sophisticated NLP tasks:

- **Semantic Search:** Find documents with similar meaning, not just matching keywords
- **Machine Translation:** Map concepts across languages
- **Sentiment Analysis:** Understand emotional tone
- **Recommendation Systems:** Suggest similar content

6.2 From 2D to High Dimensions

Real embeddings use 100-1000 dimensions because:

- Language has many subtle relationships (syntax, semantics, pragmatics)
- Higher dimensions can capture more nuanced patterns
- Multiple independent semantic axes (time, space, emotion, category, etc.)

7. Modern Extensions: Contextual Embeddings

7.1 From Static to Contextual Embeddings

Traditional word embeddings give the same vector for "bank" whether it means:

- Financial institution: "I went to the bank"
- River edge: "We sat by the river bank"

Solution: Contextual embeddings (BERT, GPT) that change based on surrounding words, providing different representations for the same word in different contexts.

Key Takeaways

- 1. Vector Foundation:** Understanding basic vector operations (dot product, cosine similarity) is crucial for modern AI.
- 2. Geometric Intuition:** Semantic relationships become geometric relationships in high-dimensional space.
- 3. Mathematical Elegance:** Simple arithmetic operations ($+$, $-$, $\cdot$) capture complex linguistic relationships.
- 4. Scalable Framework:** The same principles extend from simple word embeddings to sophisticated language models.
- 5. Distributional Power:** "You shall know a word by the company it keeps" - statistical patterns in text reveal deep semantic structures.

Conclusion

The journey from basic 2D vectors to sophisticated word embeddings illustrates how fundamental mathematical concepts can unlock powerful AI capabilities. By representing words as vectors in high-dimensional space, we enable computers to perform seemingly magical feats: understanding that "king" relates to "queen" in the same way "man" relates to "woman", or finding documents about "automobiles" when searching for "cars".

This mathematical foundation continues to evolve, from static word embeddings to dynamic contextual representations in modern transformer models. Yet the core insight remains: meaning can be captured geometrically, and similarity can be computed mathematically. Understanding these vector operations provides the key to understanding how modern AI systems process and generate human language.

As we advance toward more sophisticated AI systems, the principles explored here - vectors, similarities, and geometric relationships - remain the mathematical bedrock upon which language understanding is built.